MPHO NGWENYA

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Profession Summary

Electrical Engineering student, majoring in Computer Systems with strong foundations in programming, circuits, networking, and data science tools. Proficient in C++, Python, SQL, and hardware design, with hands-on experience using Jupyter Notebooks, RStudio, and IBM Watson Studio. Demonstrated ability to design, troubleshoot, and optimize electronic systems, power systems, and network configurations. Recognized for academic excellence and practical problem-solving in both engineering and data-driven environments.

Education

Kenneth Masekela Secondary — 2019

National Senior Certificate (NSC)

- Key Subjects: Mathematics, Physical Sciences, English (FAL)
- Achievements: Top 3 student

Sibanye-Stillwater Training Academy — 2023

Competency in Hydropower Certificate

- Acquired knowledge of hydraulic systems, safety standards, and sustainable energy operations within the mining environment.
- Learned mine safety standards, sustainable energy operations, and problem-solving in real industrial settings.

IBM — 2024

Data Science Foundation and Data Science Tools

- Gaining hands-on experience with Jupyter Notebooks, RStudio, IBM Watson Studio, and cloud-based data science and visualization tools

Vaal University of Technology (VUT) — 2026

National Diploma in Electrical Engineering – Computer Systems

- Core Courses: Electronics, Digital Systems, Networks, Engineering Programming, Operating Systems
- Achievements: Overall Average of 70%

Skills

Programming and Software

- C++
- Python
- HTML and CSS
- SQL

- MATLAB
- Multisim

Data Science Foundation and Tools

- Jupyter Notebooks
- RStudio IDE
- IBM Watson Studio
- Data Visualization Tools
- Cloud-based Data Science Environment

Circuits and Hardware Design

- Soldering
- Circuit Design
- PCB and Breadboard Prototype
- Microcontroller (Arduino)

Circuits and Hardware Design

- Power Systems Design
- Network Configuration (Router and Switch)
- IP Addressing and Subnetting
- Troubleshooting and Diagnosis

Core Competencies

- Problem Solving and Analytical Thinking
- Report Writing and Technical Communication
- Team Collaboration and Leadership
- Safety Awareness and Work Ethics
- Project Planning and Execution
- Adaptability and Continuous Learning

References

Academic Reference – Vaal University of Technology
Mr Carlos Nshimba - Program Coordinator for Computer Systems
Nshimba@vut.ac.za

Industry Reference – Sibanye Stillwater
Mr Graham Ford – Sibanye Stillwater Supervisor
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Data Science Reference – IBM
Mr CW Nduwamungu – IBM Trainer
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